

LAB-1:Use-Case Flow Specification

Problem Statement #63: Community Solar Credit Allocation Manager

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SEC:I

Use Case ID:-UC-01

Use Case Name:- Log expense

Primary Actor:- Group Traveler

Secondary Actors: Exchange Rate Service (external)

Preconditions:- The Group Traveler is logged into the app and is an active member of a trip group .
The trip has a base currency set by the trip admin.

Postconditions Success:- The expense is recorded in the group's shared ledger with the amount converted to the trip's base currency , and every affected traveler's balance is updated to reflect the new expense.

Main Success Scenario:-

1. The Group traveler selects "log expense" from the app .
2. The system displays a form requesting the amount , currency , payer/s , a short description , and the list of Travelers to split the expense between.
3. The Group Traveler enters the amount and selects the currency.
4. System validates the currency and converts the amount to the trips base currency using that day's exchange rate (include: convert currency).
5. The Group Traveler selects which Travelers the expense should be split between and confirms the entry .
6. The system validates that the split amounts sum to the total expense.
7. The system records the expense in the group's shared ledger.

8. The system recalculates the balance for every affected traveler .
9. The system sends a balance update notification to affected Travelers who have notifications enabled (notify balance update)

Alternate Flow — 4a. CURRENCY NOT SUPPORTED

- 4a. The system displays an “unsupported currency” error message .
- 4b. The Group Traveler is prompted to select a valid . supported currency .
- 4c. If the Group Traveler cancels instead of correcting the currency , the expense entry is discarded and no ledger entry is created .

